



MAL • DATA PLATFORM • DELIVERABLE 2 OF 3

Customer 360 & CAR: Entity-Relationship Diagram

The complete field-level schema for the golden Customer 360 and the CAR snapshot, drawn as one connected model, with an entity dictionary and a cardinality walkthrough for engineers who need to build against it directly.

DELIVERABLE 2 OF 3 — ERD

Companion Documents	Notation	Status	Targets
#1 Architecture #3 Identity SQL	Crow's foot (Mermaid erDiagram)	Draft for review	500K customers, Year 1

How to Read This Diagram

This diagram uses standard crow's-foot notation. A single bar means *exactly one*; a circle means *zero, optionally*; a crow's foot (three prongs) means *many*. Reading the notation at the end nearest an entity tells you the cardinality *from the other entity's point of view* – so CUSTOMER_GOLDEN || - - o{ ACCOUNT reads as "one customer owns zero or many accounts."

Solid connecting lines are enforced foreign keys inside the golden layer. CAR_SNAPSHOT is drawn as a satellite of Customer rather than a peer of the other entities because it is a derived, denormalized artifact rebuilt nightly from everything else in this diagram – nothing else holds a live foreign key into it, and it holds none into anything but Customer.

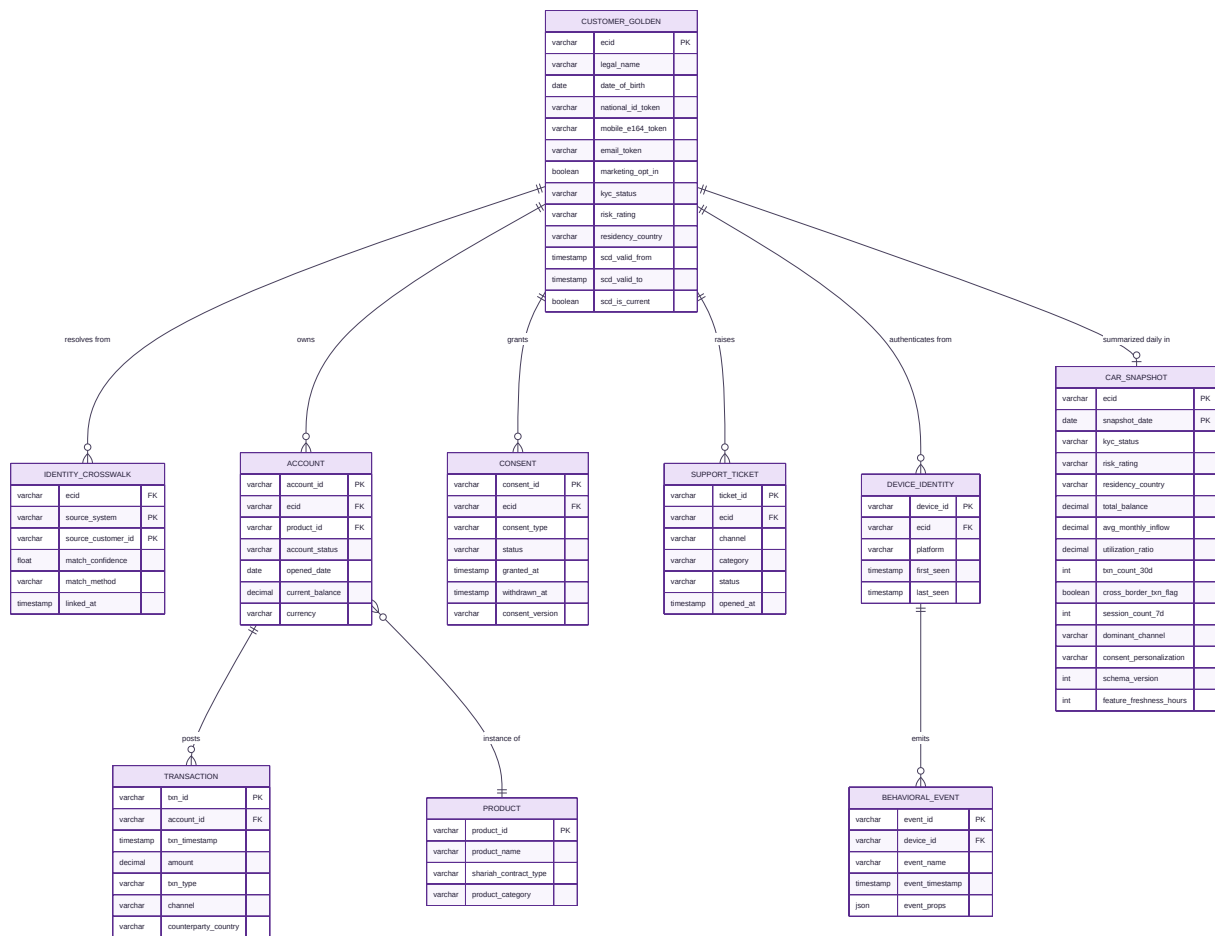


Figure 1: Golden Customer 360 core plus the CAR snapshot. Rendered from Mermaid erDiagram source; entity and column names match the SQL in Deliverable 3 exactly.



1 Entity Dictionary

Each entity below is described by what one row actually represents (its grain), its role in the schema (dimension, fact, bridge, or derived), and the fields worth calling out specifically.

1.1 CUSTOMER_GOLDEN — The Golden Identity Dimension

Grain: one row per Enterprise Customer ID (ECID) per version. **Role:** SCD Type 2 dimension – the single source of truth for who a customer is. `ecid` is generated once, at first resolution, and never changes; `scd_valid_from` / `scd_valid_to` / `scd_is_current` track history so that “what did we know about this customer on this date” is always answerable, which matters for audit and for avoiding label leakage in model training. `national_id_token`, `mobile_e164_token`, and `email_token` are tokenized, never raw, values – see Deliverable 1, Section 4. `marketing_opt_in` is the one field on this table that core banking has no equivalent for and never overwrites – per the survivorship rules in Deliverable 1, Section 2.3, it always survives from CRM.

1.2 IDENTITY_CROSSWALK — The Merge Ledger

Grain: one row per (source system, source customer ID) pair. **Role:** bridge table, and the actual mechanism behind every merge and split. This table, not `CUSTOMER_GOLDEN`, is what identity resolution writes to – an ECID’s membership is defined by which crosswalk rows point to it, which is what makes an incorrect merge reversible: correcting it is a crosswalk update and an SCD2 replay, not a destructive rewrite of the golden record itself.

1.3 ACCOUNT — The Banking Relationship

Grain: one row per account (current, savings, financing). **Role:** dimension, owned by exactly one `ecid` for Q2 (joint accounts are a Year 2 extension, modeled as an additional bridge rather than a change to this table). References `PRODUCT` rather than repeating product attributes on every account row.

1.4 PRODUCT — The Catalog

Grain: one row per product definition. **Role:** dimension. Carries `shariah_contract_type` (Murabaha, Mudarabah, Wakala, and so on) instead of an interest rate – the one place this schema visibly departs from a conventional-bank Customer 360 template, because the underlying contract structure is what both Product and Compliance actually need to reason about.

1.5 TRANSACTION — The Posted Ledger

Grain: one row per posted transaction. **Role:** fact table, additive, referencing `ACCOUNT`. Kept at transaction grain in the 360 for Support and Product to answer “what happened”; heavier aggregation for scoring happens downstream in `CAR_SNAPSHOT`, not here.

1.6 CONSENT — The Single Gate

Grain: one row per (`ecid`, consent type, version). **Role:** dimension-like control table. Every masking and personalization decision in the platform reads from this table; nothing downstream is allowed to maintain its own separate notion of whether a customer has agreed to something.

1.7 SUPPORT_TICKET — CRM Case History

Grain: one row per case. **Role:** fact table, linked to `ecid` once the originating CRM contact resolves through the crosswalk.



1.8 DEVICE_IDENTITY — The Authenticated Link to a Device

Grain: one row per Amplitude device ID. **Role:** dimension, linked to *ecid* *only* once an authenticated session confirms the device belongs to that customer – never speculatively, never by behavioral inference alone.

1.9 BEHAVIORAL_EVENT — Raw Amplitude Events

Grain: one row per event. **Role:** fact table, referencing DEVICE_IDENTITY rather than *ecid* directly, which is what makes the authenticated-binding rule in DEVICE_IDENTITY the single enforcement point for the whole behavioral stream.

1.10 CAR_SNAPSHOT — The Derived Analytic Record

Grain: one row per (*ecid*, *snapshot_date*). **Role:** derived, deliberately denormalized – rebuilt nightly from every other entity in this diagram, never updated in place. It is intentionally the widest, flattest entity in the model; see Deliverable 1, Section 5 for why that width is the point rather than an oversight. *risk_rating* and *residency_country* are carried over unchanged from CUSTOMER_GOLDEN rather than recomputed, and *feature_freshness_hours* records how stale the snapshot’s inputs were at build time – both exist so a model consuming this row never has to guess.

2 Cardinality Walkthrough

- **Customer → Identity crosswalk (one to many):** one ECID can have many source-system linkages, one per system it appears in. This is the relationship merges and splits actually operate on.
- **Customer → CAR snapshot (one to at most one, per day):** zero or one row per ECID per *snapshot_date* – drawn as one-to-one because the “many” dimension here is time, not a second live relationship.
- **Account → Product (many to one):** many accounts can be instances of the same product; the Sharia contract type lives once, on PRODUCT, never repeated per account.
- **Device identity → Behavioral event (one to many):** raw Amplitude events, aggregated nightly into the CAR’s engagement columns today, and consumed directly by the streaming layer once that exists.

WHY THIS WAY: Every foreign key in this diagram points toward *ecid*, directly or through one hop, and never the reverse. That is a deliberate constraint, not a coincidence of how the boxes happened to get drawn: it keeps identity resolution upstream of everything else in the platform, so a change to how two records get matched only ever has to propagate forward through this diagram, never backward.